AI-Powered Unified Sri Lankan Tourism Platform Using GraphRAG and LLM Orchestration

Project Proposal Report

R26-IT-050

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B.Sc. (Hons) Degree Information Technology Specializing in Information Technology

Department of Information Technology

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I. Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration. The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of Supervisor Date

Signature of Co-Supervisor Date

II. Abstract

At the present, cultural, historical, and heritage sites around Sri Lanka are visited by a large number of tourists. However, these locations are not well known to many travelers and in most cases, people visiting these places cannot identify the tourist attraction sites. Tourists will visit such sites, only to be aware of the place standing but they are not aware of the name, history, and the cultural significance of the place. The current tourism websites are primarily search engines or map-based, which is why they do not provide tourists with the opportunity to find locations with images. It is this drawback which causes tourists to be deprived of helpful information concerning the Sri Lankan heritage sites and attractions. In this study, the researchers propose an AI Landmark Recognition and Tourist Information System, which is a part of the AI-Powered Unified Sri Lankan Tourism Platform. The system enables the tourist to post an image or capture a picture of a place by use of their mobile devices. The computer vision and deep learning models will be applied to identify the tourist landmark in the image. As soon as the landmark has been recognized, the system will identify the appropriate tourist attraction. Upon identifying the landmark, a Large Language Model (LLM) will produce detailed description of the landmark. The information that will be generated will be the history of the place, cultural significance, visitor requirements, and other tourist attractions in the area. With image recognition technology and language models based on AI, the system will be able to present a smart and contextual tourism information to visitors. The proposed system aims at the creation and testing of computer vision models, which are accurate landmark recognition and their combination with the use of LLM-based information generation. This integration will enable the tourists to grasp the importance of the landmarks in Sri Lanka fast and enhance their traveling experience. In general, the system will simplify the exploration of tourism in the country by making it more interactive and more informative to both local and foreign tourists.

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1. Introduction

Sri Lanka tourism is actively going through the digital transformation speed, and it is recovering following the global pandemic. The number of visitors to the country is likely to shoot up and it is projected that Sri Lanka has the potential to have close to three million visitors by the year 2025. Together with the rise of tourism and the creation of digital technologies, various online platforms are currently assisting travelers in planning their journeys, securing accommodation, and travel suggestions. Nevertheless, the current tourism platforms continue to have a number of limitations particularly in assisting tourists to find it very easy to locate and comprehend their places of visits.

Sri Lanka also boasts of great historical and cultural features including Sigiriya and Temple of the Tooth that is a major tourist attraction each year. Nevertheless, most of the tourists encounter challenges whenever they tour such sites due to lack of knowledge on the name of the landmark or the historical and cultural significance of the place. Most of the times, tourists visit attractive sites or monuments but they lack adequate details about the sites. The available tourism applications primarily offer text-based search or map-based features that leave the user already aware of the name of the location to be searched before he or she can access information about it.

In order to solve this issue, the current study suggests the use of AI Landmark Recognition and Tourist Information System as an element of the AI-Powered Unified Sri Lankan Tourism Platform. The system enables a tourist to post or take a picture of one place through his or her mobile phone. The image obtained will then be processed with the computer vision and deep learning models to identify the tourist landmark automatically. Through the image recognition technology, the system will be able to recognize the well-known landmarks and attractions in Sri Lanka.

Once the landmark has been found, the system will rely on a Large Language Model (LLM) to provide comprehensive information about the place. The information created can consist of the history of the location, cultural value, visitor regulations, and close attractions. With this, the tourists can be able to quickly learn what the landmark is all about without manually searching and finding out its significance.

The proposed system is based on computer vision technology using language models based on AI, which offers an intelligent and interactive tourism assistant tool. The AI Landmark Recognition and Tourist Information System is meant to enhance the tourism experience by ensuring that the travelers find it easy to get to know, name, and locate landmarks in Sri Lanka. After all, this element has led to the creation of a more intelligent tourism ecosystem that allows visitors explore the heritage of Sri Lanka and its attractions in a more efficient manner.

1.1. Background & Literature Survey

Sri Lanka is a developing tourism destination and several tourists visit historical, cultural, coastal, and wildlife tourism sites throughout the State. Sri Lanka is a land of numerous popular landmarks including Sigiriya and Temple of the Tooth that seem to receive numerous local and foreign visitors annually. Nevertheless, not all of the tourists have an easy time visiting these sites since they are not aware of the name of the location or what historical significance the landmark they visit has. In most cases, tourists visit the destination without sufficient information on its cultural importance, history and other attractions.

In the present situation, the majority of tourism sites primarily offer information in either text search, travel blog or map-based navigation. Such systems require tourists to have prior knowledge of the name of the place and then they can go ahead and search the location information. Due to this shortcoming, the tourists will not be able to locate unfamiliar landmarks using pictures easily. Thus, the necessity in smart tourist systems is increasing, which are capable of identifying tourist attractions automatically and provide tourists with the necessary information.

According to the latest research in computer vision and deep learning, image recognition technologies have already managed to identify landmarks and objects in photographs with high success. Indicatively an ensemble deep learning architecture based on YOLO models in recognizing tourism landmarks on smart tourism [1]. Likewise, a multi-threshold technique of enhancing the accuracy of landmark detection was proposed [2]. These papers indicate that deep learning models are applicable to the recognition of popular tourist destinations based on images.

The utilization of Convolutional Neural Networks (CNN) to classify landmarks and recognize monuments is discussed in other works as well [3]. designed a CNN-based system that recognizes the landmarks in an image [4]. Investigated how computer vision methods can be used in a tourism information system and mobile applications. These methods emphasize the role of image classification and object recognition systems in enhancing the tourism services.

Besides computer vision models, there is also a recent research that involves the use of sophisticated models of artificial intelligence to process tourism-related data. To study the perception of tourism destinations [5]. A hybrid strategy based on the analysis of BERT, BiLSTM, and CNN. In a similar manner [6]. Research on scenic spot image image retrieval approach with the image processing.

Deep learning models were studied by other researchers [7]. In identification of heritage monuments and prediction of landmarks.

Though these studies prove the feasibility of artificial intelligence and computer vision to recognize the landmarks, most of the current systems are only oriented to image classification and detection. They do not always give an in-depth context of the identified site. When tourists recognize the landmark, they have to search independently on historical information, cultural significance, or travel advice.

Thus, intelligent tourism system involves the necessity to merge the precise identification of landmarks with the automated system of information formation. The proposed research is an AI Landmark Recognition and Tourist Information System, a system that combines computer vision models to recognize landmarks with Large Language Models (LLMs) to provide more detailed information on the identified place. Through image recognition and a language-generation system that is powered by AI, the proposed system will create a more interactive and informative experience of tourism among people visiting Sri Lanka.

1.2. Research Gap

Based on the literature review and the available systems, as illustrated in Table 1.1, it is evident that numerous research works have been conducted on computer vision and deep learning methods of identifying landmarks and monuments in images. But the majority of such solutions are concentrated on image classification or object detection. Despite the fact that these systems offer valuable input to the computer vision and tourism technology, they lack the necessary tourism-centered solution that would be able to integrate landmark recognition and smart generation of tourist information.

The current literature on smart tourism technologies is largely concerned with the accuracy of landmark detection models. As an illustration [1], presented an ensemble deep learning framework based on YOLO models to enhance the ability of landmarks recognition in smart tourism settings. In the same manner, the method [2], has proposed a multi-threshold enhancement and selective ensemble approach to improve the landmark recognition performance. Although the performance of these methods is good in the recognition of landmarks, the main focus of most of them is on enhancing the accuracy of the model and detection efficiency other than actual provision of meaningful tourism information to the travelers.

The use of Convolutional Neural Networks (CNN) and image processing techniques in the process of landmark and monument identification have been explored by other research studies. As an example [3], constructed a CNN-based image landmark recognition model in image datasets [4], investigated the application of computer vision technique in tourism information systems in mobile devices. Even though these systems are good in recognizing landmarks of images, they remain primarily technical recognition systems that lack contextual explanations and historical insights regarding the identified tourist sites.

Moreover, artificial intelligence to analyze tourism images and recognize heritage is discussed in a number of researches. A hybrid framework of BERT, BiLSTM, CNN, and attention to understand the perception of tourist destinations [5]. Equally experimented on scenic spot image retrieval in terms of image processing technology [6], tested deep learning architecture to identify heritage monuments [7]. Although these studies prove that AI has a potential in tourism applications, the majority of them deal with image retrieval, classification, or behavior analysis instead of providing tourists with rich and informative descriptions.

Moreover, there are certain studies including Ai et al. Investigate the image recognition and tourism data analysis integration [8]. The systems however are designed to analyze and monitor data and not to help the tourist locate landmarks directly and learn their cultural and historical significance.

Hence, it is evident that a vast majority of the current research is dedicated to landmark recognition and image classification with minimal studies on the combination of computer vision models and sophisticated AI language models to give contextual tourist information. Even now, it is still required to have a single intelligent tourism system that is able to recognize landmarks based on pictures and provide meaningful descriptions concerning the location without human input.

The proposed research will help fill the above-mentioned gap and present an AI Landmark Recognition and Tourist Information System. The suggested system will combine image recognition based on deep learning and Large Language Models (LLM) to automatically recognize tourist landmarks and provide more detailed contextual information regarding the landmark. The system will enhance the tourism experience of the residents to learn more about the heritage sites in Sri Lanka, and this is possible through the integration of these technologies into a cohesive system to support tourists achieve a more interactive and informative tourism experience.

The following are the primary features that will be compared to my research component.

Feature	Landmark Recognition	Image Classification	Image Classification	Image Classification	Image Classification
Research A [1]	✓	✓	✗	✗	✗
Research B [2]	✓	✓	✗	✗	✗
Research C [3]	✓	✓	✗	✗	✗
Research D [4]	✓	✓	✗	✗	✗
Research E [5]	✗	✓	✗	✗	✗
Research F [6]	✓	✓	✗	✗	✗
Research G [7]	✓	✓	✗	✗	✗
Proposed system	✓	✓	✓	✓	✓

Table 1. 1: Table of Research Gap

1.3. Research Problem

Sri Lanka is a rapidly growing country in terms of tourism and most of the tourists visit various historical, cultural, and heritage sites in the country. Sri Lanka boasts some of the well-known tourist destinations including Sigiriya, temple of the tooth, and other monuments and heritage sites. Nevertheless, most tourists have had a challenge of recognizing these landmarks when they visit them, particularly when they are on their own with no tour guide accompanying them.

The majority of tourists use either guide books, online search or map based tourism applications to find details about the tourist attraction. The systems are however unable to show any details about a location unless the person who is searching has already known the name of that location. Tourists will observe a monument or a landmark in most cases, without necessarily knowing the name of the site or its historical and cultural significance. Consequently, they will not acquire valuable information on heritage sites and attractions of Sri Lanka.

Despite the development of computer vision and deep learning technologies that are used to identify objects and detect objects in images, most of the current systems are only oriented at object recognition or image classification. The use of these systems lacks a comprehensive description and tourism information following the discovery of the landmark. Thus, tourists will still have to search manually to get more information about the place through other platforms.

Moreover, most of the tourism applications lack intelligent support which enables the tourist to locate the landmarks with ease with assistance of pictures taken on their mobile devices. In the absence of such technology, tourists might find it hard to visit new destinations and comprehend the meaning of historical sites that they are visiting.

Thus, the proposed study is intended to create an AI Landmark Recognition and Tourist Information System enabling tourists to find a landmark with the help of the images taken with their smartphones. It will be based on the computer vision models that will identify the landmark and then an Large Language Model (LLM) will provide all the detailed information about the identified location, its history and cultural relevance, information on how to visit it and places of interest around the location. The given system will utilize image recognition and AI-based information generation to enhance the tourism experience and make guests more aware of the landmarks and heritage sites of Sri Lanka

1.4 Research Questions

For the research gaps and research problem identified, the research questions for the study are as follows:

1. **Automated Landmark Recognition**: How can images captured using a smartphone camera be used with computer vision and deep learning models to automatically detect and recognize tourist landmarks?

2. **Accurate Landmark Classification**: How can Artificial Intelligence and deep learning techniques be used to accurately classify and identify different tourist attractions and heritage landmarks in Sri Lanka?

3. **AI-Based Tourist Information Generation:** How can Large Language Models (LLMs) be used to generate meaningful tourist information such as the history, cultural significance, and visitor guidance of identified landmarks?

4. **Contextual Tourism Assistance:** What potential does the integration of computer vision and AI language models hold for providing contextual explanations and intelligent tourism assistance to visitors exploring unfamiliar locations?

5. **Mobile-Based Tourism Support:** How can a mobile-based AI system be designed to allow tourists to easily capture images of landmarks and instantly receive detailed information about the place and nearby attractions?

2. Objectives

2.1. Main Objective

The main objective of this research is to conceptualize and design a Landmark Recognition and Tourist Information System using Artificial Intelligence technology that can automatically identify tourist landmarks based on images and provide information on the location using Artificial Intelligence technology.

The proposed Landmark Recognition and Tourist Information System will be based on computer vision technology that recognizes tourist landmarks based on images taken by a smartphone's camera. Once the landmark is recognized by the proposed system, a Large Language Model (LLM) will be used to generate relevant information on the location.

The proposed Landmark Recognition and Tourist Information System has the potential to greatly enhance the tourism experience by enabling tourists to easily identify landmarks and understand the importance of the location without any prior knowledge on the location. The proposed Landmark Recognition and Tourist Information System has the potential to provide a smart and interactive platform for tourists to explore the heritage locations in Sri Lanka.

2.2. Specific Objectives

1. Develop an AI-Based Landmark Recognition System

The first specific objective is to design and implement a computer vision–based system that can automatically recognize tourist landmarks using images captured by tourists.

• Image-Based Landmark Detection: Develop a deep learning model that can analyze images captured from smartphone cameras and identify well-known tourist attractions and heritage sites in Sri Lanka.

• Landmark Classification: Train the model using a dataset of Sri Lankan landmarks so the system can correctly classify different tourist locations such as historical monuments, cultural sites, and famous attractions.

• Mobile Image Processing: Ensure that the system can process images uploaded from mobile devices and return recognition results quickly for tourists in real time.

2. **Implement AI-Based Tourist Information Generation**

The second objective is to develop an intelligent information generation system that can provide meaningful explanations about the recognized landmarks.

• LLM-Based Information Generation: Use Large Language Models (LLMs) to automatically generate detailed information about the identified landmark, including its history, cultural significance, and importance.

• Contextual Tourism Explanation: Provide contextual explanations to help tourists understand the background of the location they are visiting.

• Visitor Guidance Information: Generate useful visitor information such as visiting guidelines, travel tips, and nearby attractions that tourists may also want to explore.

3. **Integrate Computer Vision with AI Language Models**

The third objective is to integrate the landmark recognition model with the LLM-based information system to create a complete intelligent tourism assistant.

• Recognition-to-Information Pipeline: Design a system workflow where the recognized landmark from the image recognition model is automatically sent to the LLM to generate related information.

• Automated Tourism Assistance: Ensure the system provides a seamless experience where tourists can upload a photo and instantly receive accurate landmark identification and detailed information.

4. **Develop a Mobile-Friendly Tourism Assistance Platform**

The fourth objective is to design the system in a way that it can easily be used by tourists through mobile devices.

• User-Friendly Image Upload Interface: Allow tourists to capture or upload images easily through a simple mobile interface.

• Real-Time Response: Provide quick responses so tourists can immediately learn about the landmark they are visiting.

• Interactive Tourism Experience: Improve the overall tourism experience by helping visitors explore and understand Sri Lanka's historical and cultural sites in an interactive way.

3. Methodology

The AI Landmark Recognition and Tourist Information System is developed using the methodology of creating an intelligent tourism assistance framework, which integrates computer vision technology and Large Language Models (LLMs). As opposed to the conventional tourism information system which requires manual search or text queries, the proposed system gives an opportunity to the tourist to locate landmarks directly by using images they still have on their smartphones.

The procedure is an image-based landmark recognition methodology. Under this method, tourists will be able to make a snapshot or post a picture of a destination using a mobile application. The uploaded image is processed with the help of the computer vision methods and deep learning models, which identify the significant visual elements of the landmark. The trained landmark recognition model analyzes these features to establish the right tourist attraction.

The system is based on an image classification model that is trained with deep learning, e.g. Convolutional Neural Networks (CNN) or object detecting models, on a dataset of established tourist landmarks in Sri Lanka. The model will be able to visualise the patterns of various landmarks and by comparing the uploaded image with the trained data, it will be able to determine the location.

Upon identifying the landmark, the system forwards the name of the identified landmark to the Tourist Information Generation Module, which is an AI-based module. A Large Language Model (LLM) in this module processes the name of the landmark and provides a comprehensive description of the place. The information created entails the history of the landmark, cultural importance, visitor regulations together with adjoining tourism sites which could be of benefit to the travelers.

Once the information has been generated, the results are presented to the tourist on the mobile application interface. The system gives the name of the identified landmark and the elaboration of the landmark and the tourist can just grasp the meaning of what place he or she is visiting.

Lastly, the system logs user interactions and recognition outcome in a data monitoring and evaluation layer. The layer is used to analyze the performance of the system, better the accuracy of landmark recognition, and advance the quality of the information introduced by AI with time.

This is a systematic approach to allow the system to offer prompt, intelligent and interactive tourism referral services, allowing visitors to easily recognize landmarks and learn more about the cultural and historical background of Sri Lanka.

3.1. System Architecture Diagram

The AI Landmark Recognition and Tourist Information System is designed using a multi-tier intelligent architecture that integrates mobile image capture, computer vision processing, artificial intelligence based information generation, and tourism information delivery. The system processes images captured by tourists and automatically identifies landmarks while providing contextual tourism information. The architecture is organized into five main functional layers as shown in the system architecture diagram.

A. Layer 1: User Interface Layer

This is the entry point of the system where the tourist interacts with the mobile application. Through the mobile interface, users can capture images of landmarks or upload existing images from their device gallery. The interface layer collects user inputs and sends them to the processing engine.

The main inputs collected in this layer include: Captured image of a tourist attraction, GPS location data of the user, User interaction requests through the mobile application

This layer ensures a simple and user friendly experience for travelers when interacting with the system.

B. Layer 2: Image Processing and Feature Extraction Layer

In this layer, the captured image is preprocessed and analyzed using computer vision techniques. The goal of this stage is to extract important visual features from the image that can be used for accurate landmark recognition.

Key processes performed in this layer include: Image preprocessing such as resizing, normalization, and noise reduction, Feature extraction using computer vision algorithms, Preparation of image data for deep learning models

This layer improves the quality of the input image and prepares it for the landmark recognition stage.

C. Layer 3: Landmark Recognition Engine

This layer represents the core recognition component of the system where deep learning models are used to identify landmarks from the processed image.

The recognition process is performed using: Convolutional Neural Networks (CNN) or object detection models, A trained landmark database containing images of Sri Lankan tourist attractions, Image classification algorithms to match the input image with known landmarks

D. Layer 4: AI Information Generation Layer

After identifying the landmark, the system generates detailed tourism information using artificial intelligence. This layer uses a Large Language Model (LLM) to generate contextual explanations about the recognized location.

The generated information may include: historical background of the landmark, cultural significance and importance, visitor guidelines and rules, nearby tourist attractions and points of interest

This layer transforms raw recognition results into meaningful tourism knowledge for the traveler.

E. Layer 5: Information Presentation and Monitoring Layer

The final layer delivers the generated information to the tourist through the mobile application interface. The system presents the recognized landmark name along with AI generated descriptions and relevant tourism insights.

Key outputs produced by the system include: Recognized landmark name, AI generated tourist information, cultural and historical explanations, nearby attraction recommendations

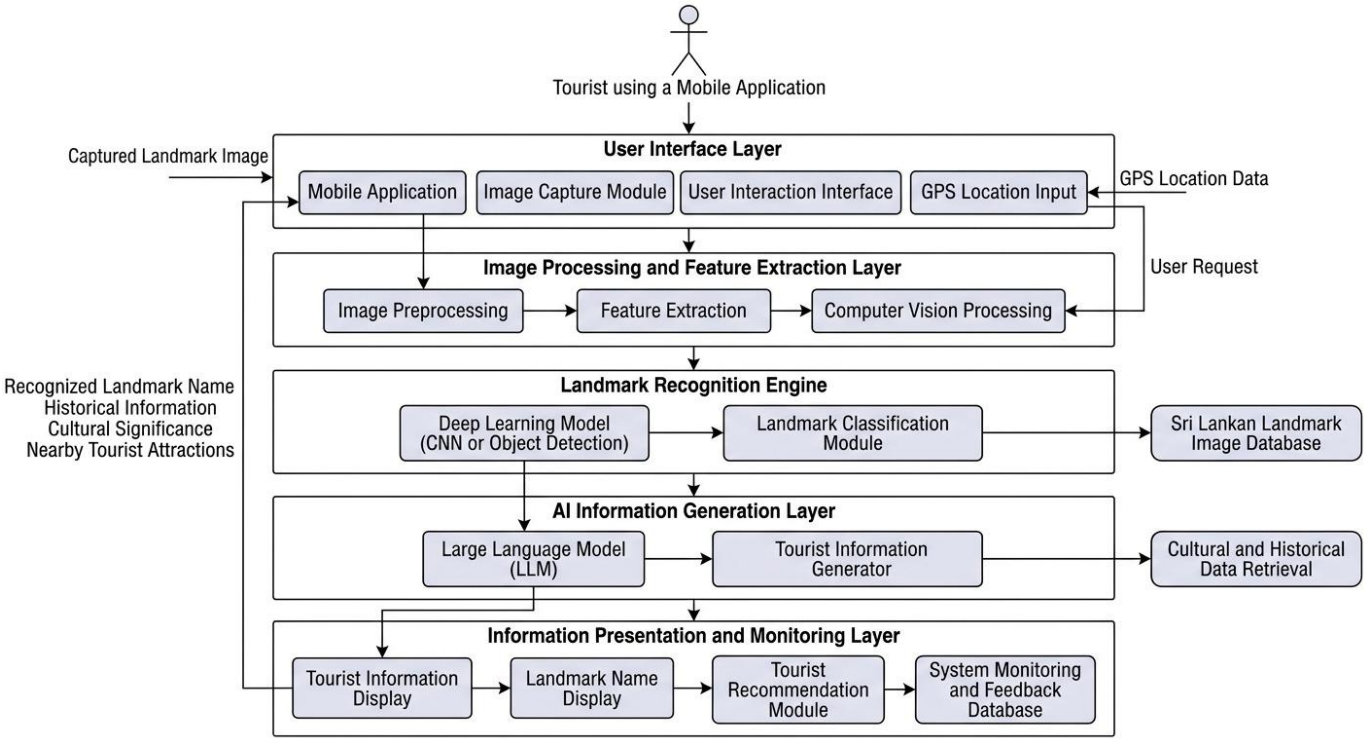


Figure 1 : System Architecture Diagram

3.2. Technologies to be used

Backend & API Management:

• Node.js: Used as the backend runtime environment to handle system requests and manage communication between the mobile application, AI models, and the database.

• Express.js: Used to build RESTful APIs that allow the mobile application to send images to the server and receive landmark recognition results and tourist information.

AI & Computer Vision Core:

• Deep Learning Models (CNN / YOLO): Used for image processing and landmark recognition. These models analyze uploaded images and identify tourist landmarks based on trained datasets.

• Large Language Model (LLM): Used to generate detailed tourist information about the recognized landmark, including history, cultural importance, and visitor information.

• Google AI Studio / Gemini API: Can be used to generate contextual explanations and tourism information based on the identified landmark.

Database & Storage:

• MongoDB Atlas (NoSQL): Used to store landmark datasets, tourist information records, user interaction data, and system logs. The NoSQL structure allows flexible storage of different types of tourism-related information.

• Cloud Storage: Used to temporarily store uploaded images for processing and model training purposes.

Frontend & Mobile Application:

• React Native: Used to develop the mobile application for tourists. It allows users to capture or upload images of landmarks and view the generated tourist information.

• Next.js: Used to develop a web-based dashboard for administrators to monitor system performance, manage landmark datasets, and analyze system usage.

AI Model Development Tools:

• Python: Used for training deep learning models for landmark recognition.

• TensorFlow / PyTorch: Used to build and train computer vision models that recognize landmarks from images.

3.3. Work Breakdown Structure

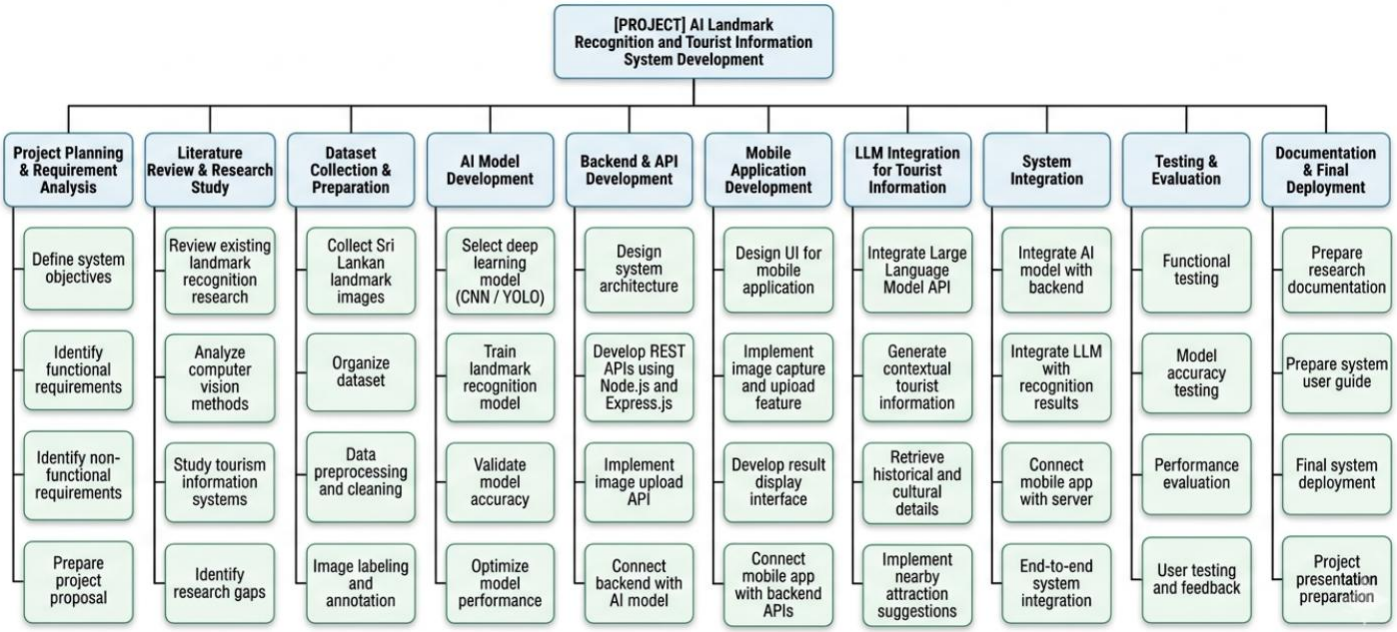


Figure 2: Work Breakdown Structure

4. Project Requirements

4.1. Functional Requirements

- Image Upload and Capture: The system should allow tourists to capture a photo using the smartphone camera or upload an image of a landmark through the mobile application.
- Landmark Recognition: The system should analyze the uploaded image using computer vision models and identify the tourist landmark accurately.
- Landmark Classification: The system should classify different types of tourist attractions such as historical monuments, cultural sites, and famous landmarks in Sri Lanka.
- AI Tourist Information Generation: After recognizing the landmark, the system should automatically generate detailed tourist information including history, cultural significance, visiting details, and nearby attractions using a Large Language Model (LLM).
- Display Tourist Information: The mobile application should display the recognized landmark name together with the AI-generated explanation and related tourist information.

4.2. Non-Functional Requirements

- **High Recognition Accuracy:** The system should maintain high accuracy in identifying landmarks from images.
- **Fast Processing Time:** The system should process uploaded images quickly and return results in a short time.
- **Reliable System Performance:** The system should provide stable performance even when many users access the platform simultaneously.
- **Secure Data Handling:** User-uploaded images and system data should be handled securely to protect user privacy.
- **User-Friendly Interface:** The mobile application should provide a simple and easy-to-use interface so tourists can easily capture images and receive information.

4.3. System Requirements

This section describes the minimum hardware and software requirements for the successful development and operation of the AI Landmark Recognition and Tourist Information System.

A. Hardware Requirements

● Server-Side Infrastructure: A cloud-based server with at least 8 GB RAM, multi-core CPU, and GPU support for processing deep learning models and AI-based information generation.
● Client-Side Devices: Smartphones with the following features: Camera for capturing landmark images, GPS module for location-based tourism suggestions, Internet connectivity (4G/5G/Wi-Fi), Minimum 3 GB RAM for smooth mobile application performance
● AI Processing Infrastructure: Servers capable of running deep learning models for landmark recognition and AI-based tourist information generation.

B. Software Requirements:

● Operating Systems: Linux (Ubuntu 20.04+) for server deployment , Android 10.0+ and iOS 14.0+ for mobile application compatibility
● Development Stack: Node.js with Express.js for backend API development, React Native for cross-platform mobile application development, Next.js for web-based system management dashboard

● Database Management: MongoDB Atlas (NoSQL) for storing landmark data, tourist information, and system logs, Cloud storage for temporarily storing uploaded images
● AI Processing Engine: Python-based AI services for landmark recognition model development, TensorFlow or PyTorch for training deep learning models, Large Language Model (LLM) APIs for generating tourist information about recognized landmarks

4.4. Test Cases

1. Landmark Recognition Accuracy: Verify that the system correctly identifies tourist landmarks from uploaded images with high accuracy.

2. Image Processing Performance: Test whether the system processes uploaded images quickly and returns results within an acceptable response time.

3. AI Tourist Information Generation: Ensure the system generates correct and relevant tourist information for the identified landmark.

4. Mobile Application Functionality: Validate that tourists can successfully capture or upload images and receive recognition results through the mobile application.

5. System Integration Testing: Confirm that all system components (mobile app, backend, AI recognition model, and LLM information module) work together correctly.

5. Gantt Chart



Figure 3: Gantt Chart

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